

Software Development Fundamentals (SDF) – I

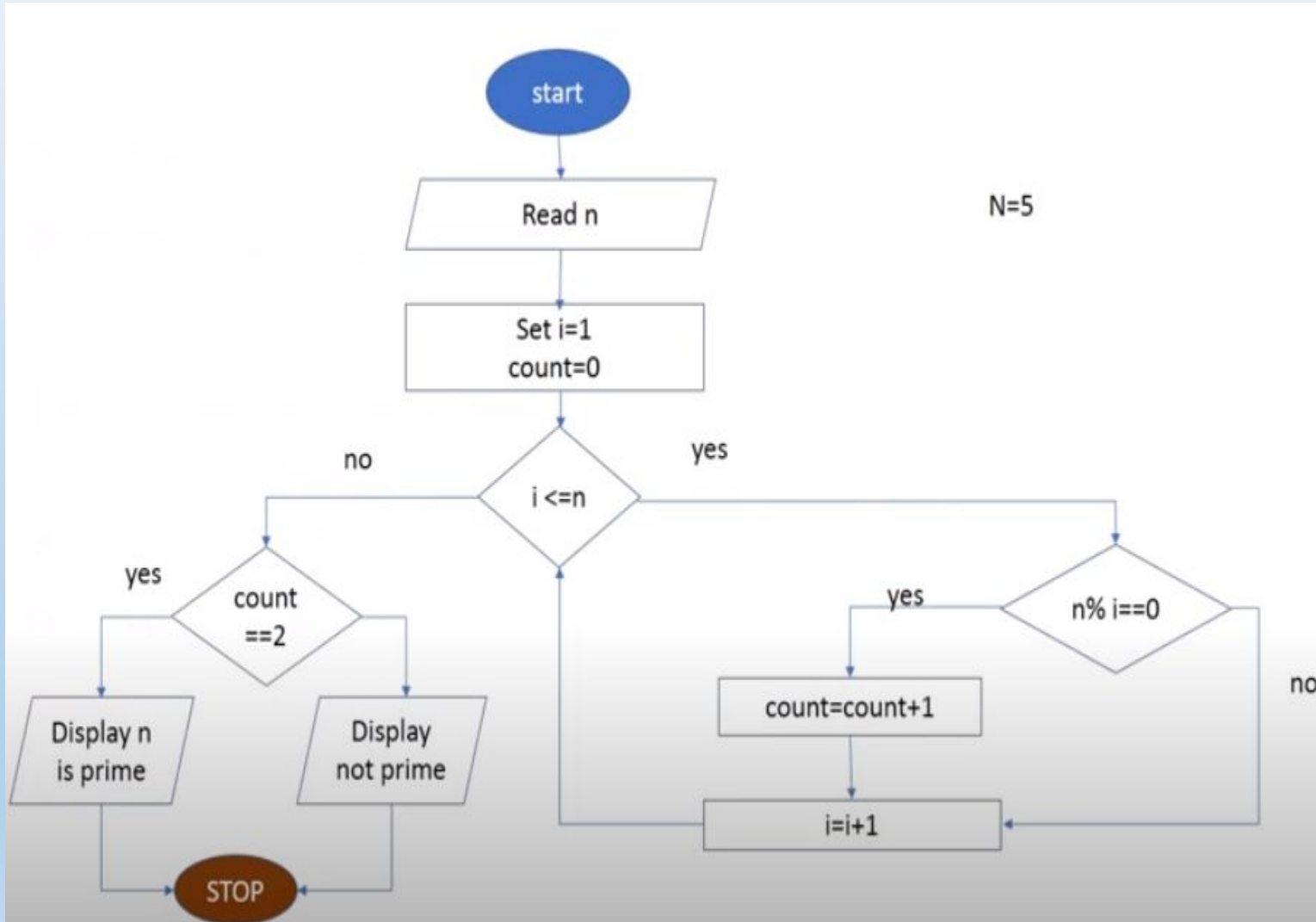
ODD 2023



Drawing flow chart and writing C programs

**Jaypee Institute of Information Technology (JIIT),
Noida**

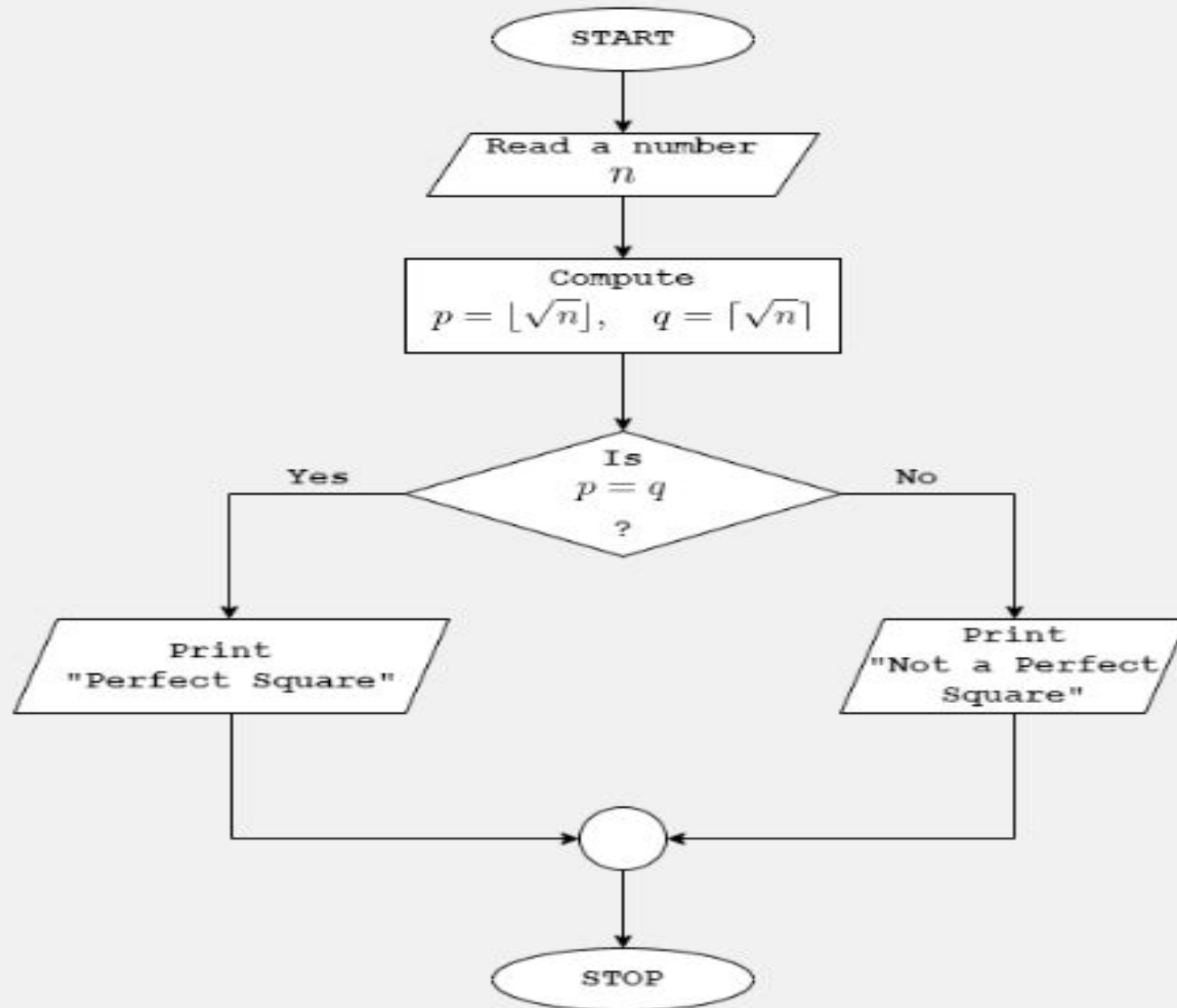
Flow Chart of Prime number



C program of Prime number

```
#include<stdio.h>
int main()
{
    int n,i, count=0;
    clrscr();
    printf("enter n value\n");
    scanf("%d",&n);
    for(i=1;i<=n;i++)
    {
        if(n%i==0)
        {
            count=count+1;
        }
    }
    if(count==2)
        printf(" %d is prime",n);
    else
        printf(" %d is not prime",n);
}
```

Flow Chart to check perfect square



C program to check perfect square

```
#include <stdio.h>
#include <math.h>

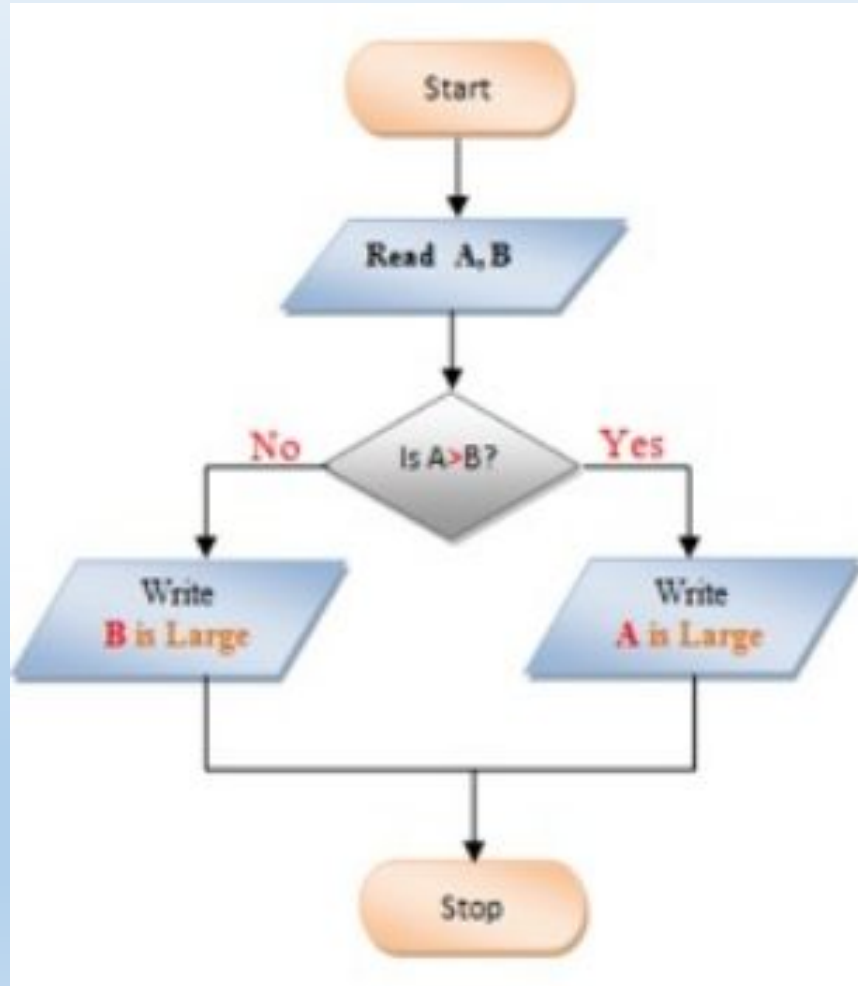
int main() {
    // declare variables
    int n;

    // take input
    printf("Enter a number: ");
    scanf("%d", & n);

    // check if ceiling and floor value
    // of square root of n are same or not
    // If the values are same, then n is a perfect square
    if (ceil((double) sqrt(n)) == floor((double) sqrt(n)))
        printf("%d is a perfect square", n);
    else
        printf("%d is not a perfect square", n);

    return 0;
}
```

Flow Chart to find the largest of two numbers



C program to find the largest of two numbers

```
#include<stdio.h>

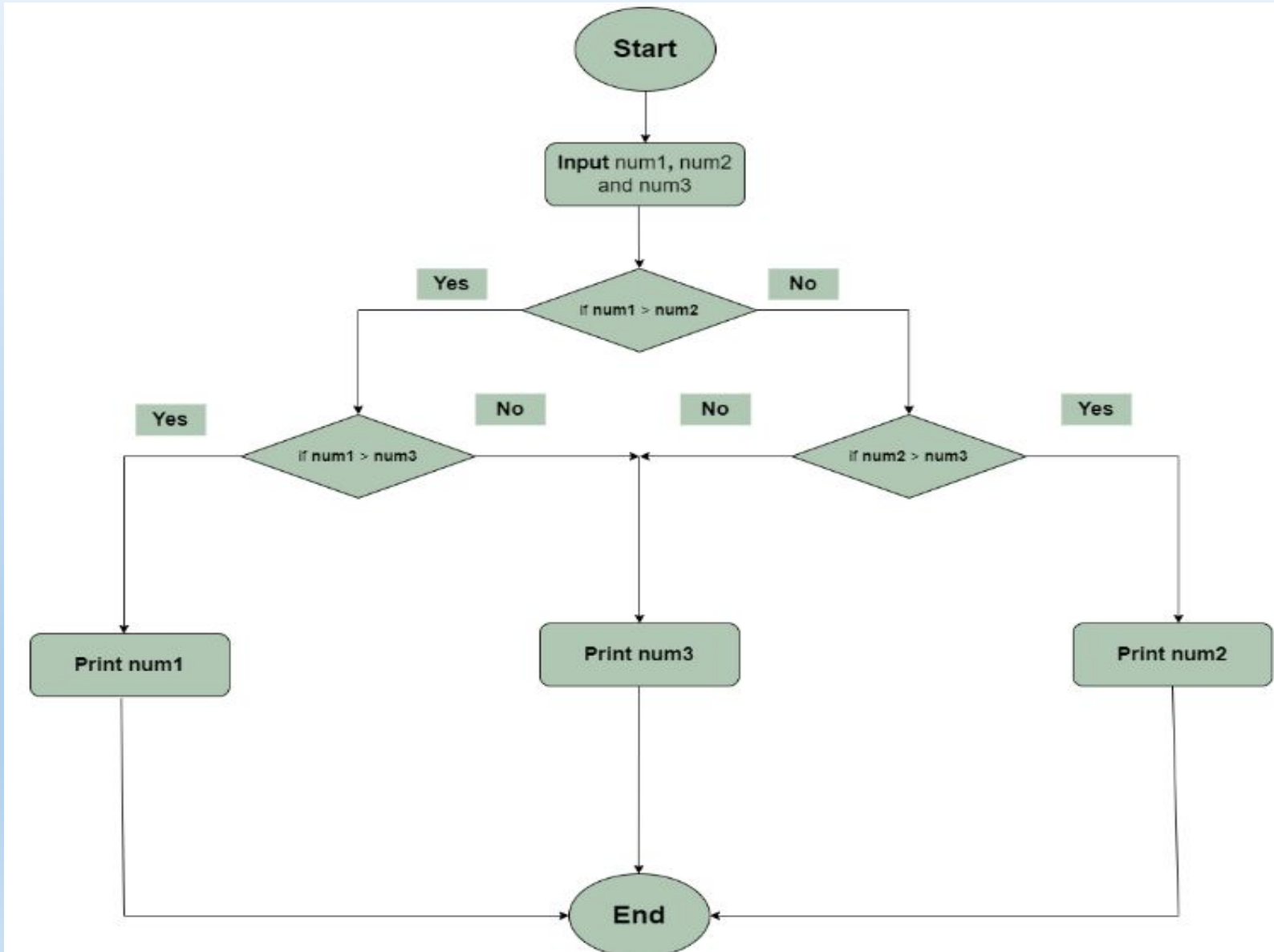
int main()
{
    int A, B;

    printf("Enter values of A, B: ");
    scanf("%d %d", &A, &B);

    if (A>B)
        printf("A is Larger");
    else
        printf("B is Larger");

    return 0;
}
```

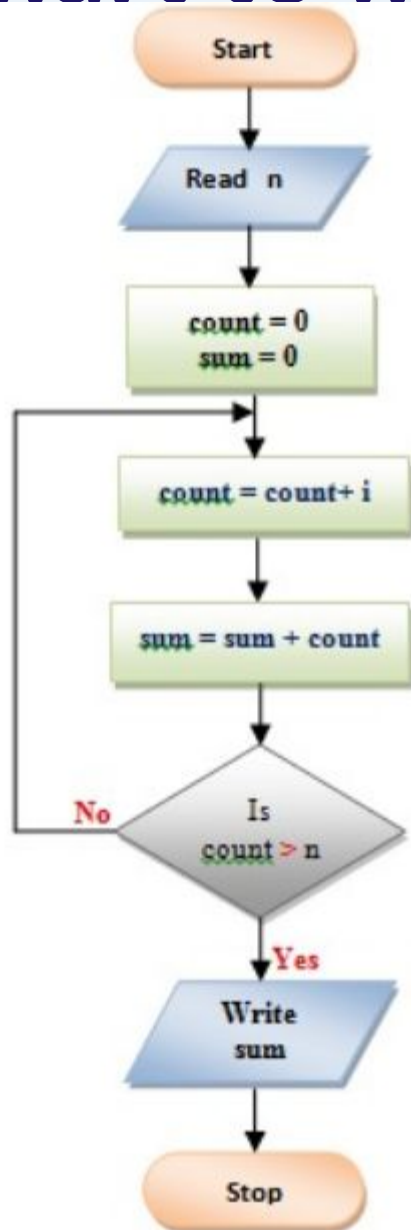
Flow Chart to find the largest of three numbers



C program to find the largest of three numbers

```
#include <stdio.h>
int main()
{
    int num1, num2, num3;
    //Get and store value in num1
    printf(" Enter the number1 = ");
    scanf("%d", &num1);
    //Get and store value in num2
    printf("\n Enter the number2 = ");
    scanf("%d", &num2);
    //Get and store value in num3
    printf("\n Enter the number3 = ");
    scanf("%d", &num3);
    if (num1 >= num2)
    {
        if (num1 >= num3)
        {
            printf("\n %d is the largest number.\n", num1);
        }
        else
        {
            printf("\n %d is the largest number.\n", num3);
        }
    }
    else
    {
        if (num2 >= num3)
        {
            printf("\n %d is the largest number.\n", num2);
        }
        else
        {
            printf("\n %d is the largest number.\n", num3);
        }
    }
    return 0;
}
```

Flow Chart to find sum of n natural numbers



C program to find sum of n natural numbers

```
#include<stdio.h>

int main()
{
    int n, i, sum;

    sum = 0;

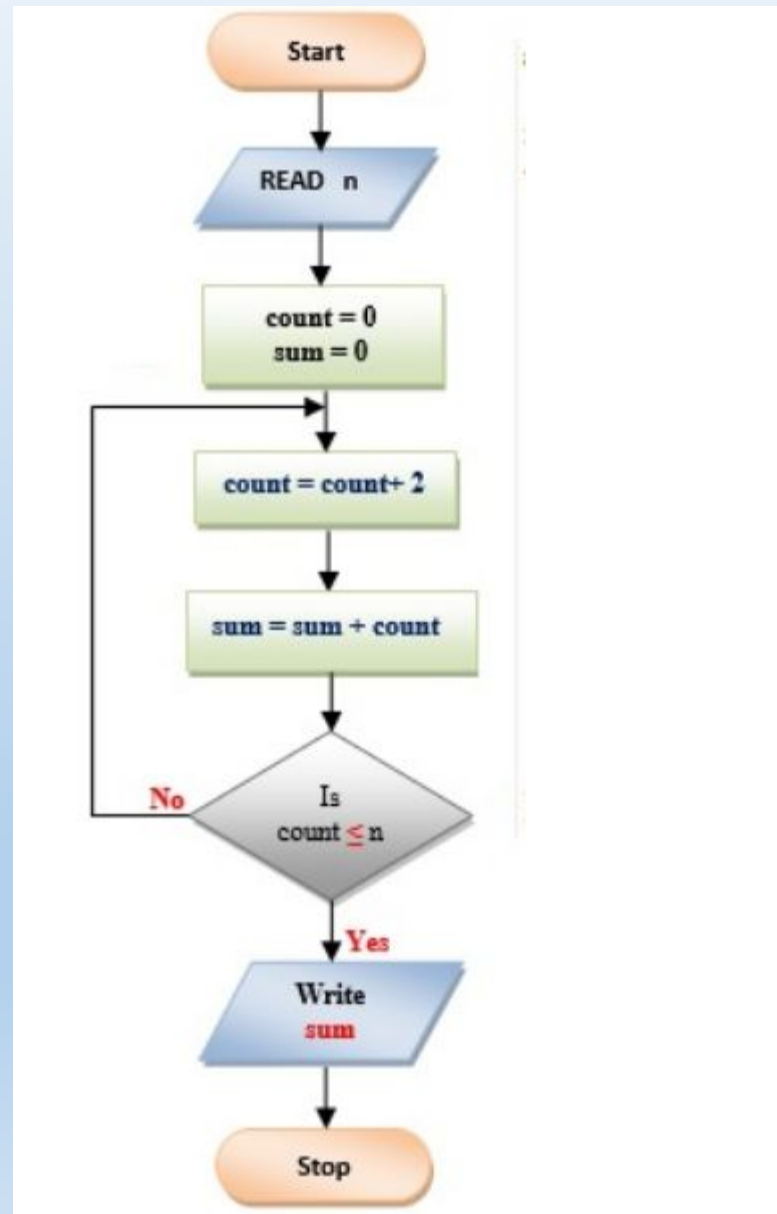
    printf("Enter value of n: ");
    scanf("%d", &n);

    for(i=1; i<=n;i++)
    {
        sum = sum + i;
    }

    printf("Sum of n natural numbers is: %d", sum);

    return 0;
}
```

Flow Chart to find sum of all even numbers up to n



C program to find sum of all even numbers up to n

```
#include<stdio.h>

int main()
{
    int n, count, sum;

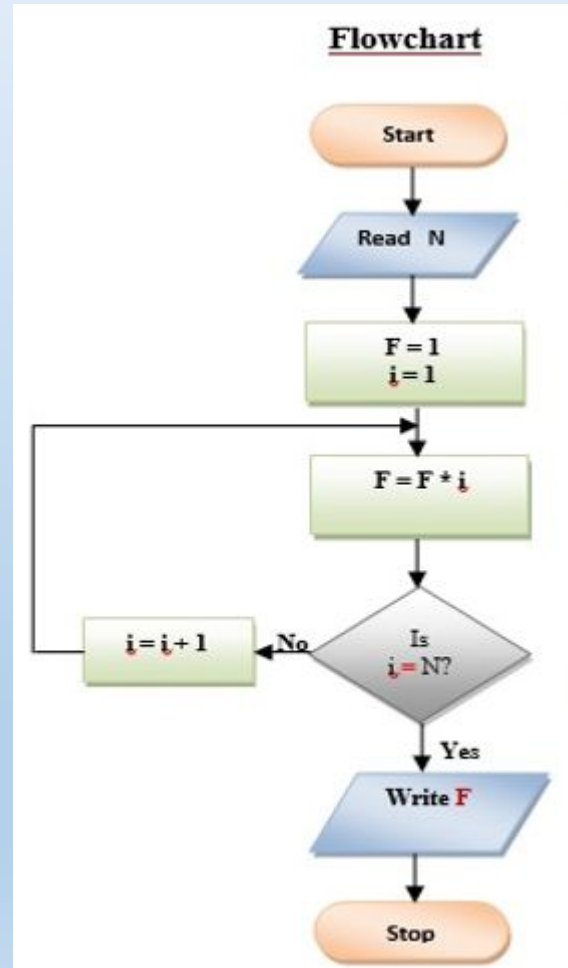
    printf("Enter value of n: ");
    scanf("%d", &n);

    count = 0;
    sum = 0;
    while (count<=n)
    {
        sum = sum + count;
        count = count + 2;
    }

    printf("Sum of even numbers upto n is: %d", sum);

    return 0;
}
```

Flow Chart to compute factorial N



C program to compute factorial N

```
#include<stdio.h>

int main()
{
    int N, F = 1, i;

    printf("Enter value of N: ");
    scanf("%d", &N);

    for(i=1; i<=N;i++)
    {
        F = F * i;
    }

    printf("Factorial of N is: %d", F);

    return 0;
}
```

- [https://www.tutorialspoint.com/cprogramming/c_relational_operators.htm\](https://www.tutorialspoint.com/cprogramming/c_relational_operators.htm)
- <https://www.programiz.com/c-programming/c-operators>

The End